

Reading the Rules Written in Disease: Kolmogorov-Arnold Networks on Clinical Manifolds

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Abstract

We investigate whether Kolmogorov-Arnold Networks (KANs) can both *classify* and *symbolically explain* disease boundaries on clinical data manifolds. Across five UCI benchmarks — heart disease (303 samples, 13 features), breast cancer (569×30), Pima Indians diabetes (768×8), Parkinson’s voice (195×22), and dermatology skin classification (366×34) — we compare three models in 5-fold cross-validation: a zero-parameter geometric baseline (**ManifoldModel**), a KAN trained on all ambient features (**KAN-raw**), and a KAN trained on the d^* -dimensional intrinsic PCA coordinates discovered by the WaveRider manifold stack (**KAN-pca**).

Three findings emerge. **(1) Accuracy:** ManifoldModel wins or ties on four of five datasets despite having zero learned parameters, confirming that the data manifold geometry alone encodes the discriminative signal. **(2) AUC-ROC:** KANs produce calibrated risk scores that ManifoldModel cannot; AUC gaps of 11–16 pp on Parkinson’s and diabetes show that continuous probability output is clinically indispensable for screening and risk stratification. **(3) Symbolic regression:** On the heart disease dataset, a one-layer KAN trained on $d^*=9$ intrinsic coordinates converges to a single linear formula, $\text{logit}(p) \approx 2.557 \cdot \text{PC1} - 0.035$, where PC1 loads on exercise-induced ST depression (oldpeak), maximum heart rate (thalach), and ST-slope — exactly the trio measured by clinical stress testing. The network recovers a 60-year-old cardiology finding from raw biomarker data without domain supervision. These results suggest a two-model clinical deployment architecture: ManifoldModel for primary hard classification, KAN-pca for continuous risk stratification and symbolic audit.

1. Introduction

Clinical AI has a well-documented generalization problem. Models trained on tabular biomedical data routinely exhibit strong in-sample performance but degrade across institutions, demographic subgroups, and equipment cohorts (Cheplygina et al., 2019). The root cause is usually the same: the model fits high-dimensional noise rather than the low-dimensional signal. For a dataset with n ambient features and intrinsic dimension d^* , the ratio $(n - d^*)/n$ measures the noise fraction. Across the five datasets studied here this ratio ranges from 25% to 73%, meaning that a naive classifier operating in ambient space is fighting 25–73% noise by construction.

In prior work (Suchanek, 2026), we showed that restricting classification to the d^* -dimensional tangent subspace of the data manifold — using a zero-parameter geometric classifier called **ManifoldModel** — achieves accuracy that matches or exceeds trained neural networks on several clinical benchmarks. ManifoldModel makes hard binary decisions: a majority vote over the k nearest tangent-space neighbors. It is accurate, fast, and

interpretable in its architecture, but it cannot produce a continuous risk score, and it cannot tell us *which algebraic function* separates the disease classes.

This paper addresses both limitations using Kolmogorov-Arnold Networks (KANs) (Liu et al., 2024). The KAN theorem guarantees that any continuous multivariate function can be decomposed into a sum of single-variable functions. For clinical classification, this means: if the disease decision boundary is smooth — and manifolds are smooth by definition — then it can be symbolically identified as a sum of interpretable 1-D curves. When those curves reduce to known functions (linear, Gaussian, sigmoid), we obtain a closed-form formula that a clinician or regulatory body can audit directly.

Our experimental design has three arms per dataset:

1. **ManifoldModel**: zero-parameter, purely geometric baseline.
2. **KAN-raw**: KAN trained on all ambient features.
3. **KAN-pca**: KAN trained on the d^* intrinsic PCA coordinates discovered by the WaveRider local-PCA pipeline at variance threshold $\tau = 0.90$.

After 5-fold cross-validation we run two-phase symbolic regression on the full training set for each dataset to extract a human-readable formula for the decision boundary.

Contributions.

- Benchmark evidence that ManifoldModel accuracy generalizes across five heterogeneous clinical datasets (Section 5).
- Evidence that KAN AUC-ROC substantially exceeds ManifoldModel AUC on high-noise datasets, establishing complementary roles for the two models (Section 5).
- Symbolic recovery of the heart disease decision boundary as a linear function of exercise-induced cardiac stress biomarkers, without domain supervision (Section 6).
- A two-model deployment architecture for clinical AI combining the strengths of both approaches (Section 7).

2. Background

2.1. The Manifold Hypothesis in Clinical Data

The manifold hypothesis (Bengio et al., 2013) holds that high-dimensional data concentrates near a low-dimensional smooth submanifold of ambient space. For clinical tabular data this is empirically verifiable: local PCA at a variance threshold $\tau = 0.90$ identifies a stable intrinsic dimension d^* for each dataset (Table 1).

The WaveRider ManifoldModel exploits this structure by (a) building a k -NN graph of training points, (b) computing an orthonormal tangent basis at each node via local PCA, and (c) classifying queries by majority vote over their k nearest tangent-space neighbors (Suchanek, 2026). The classifier has zero learned parameters; its entire “model” is the stored graph and local bases, which together encode the manifold geometry.

2.2. Kolmogorov-Arnold Networks

Kolmogorov’s 1957 superposition theorem states that any continuous function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ can be written as

$$f(x_1, \dots, x_n) = \sum_{q=0}^{2n} \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(x_p) \right), \quad (1)$$

where each $\phi_{q,p}$ and Φ_q is a continuous single-variable function (Kolmogorov, 1957). KANs (Liu et al., 2024) operationalize this theorem by replacing fixed neuron activations with *learnable spline functions on the edges*. Each edge ϕ_{ij} between input i and hidden neuron j is a B -spline with trainable coefficients. After convergence, an automatic symbolic regression step (`auto_symbolic`) fits known functions (polynomial, Gaussian, sigmoid, sinusoid) to each learned spline by least squares, recovering a closed-form expression when the spline is well-approximated by a standard function.

The key coupling with our manifold framework: we know the d^* intrinsic dimensions before training. Feeding those coordinates to a KAN gives the network the signal subspace directly, reducing the fitting problem from n -dimensional to d^* -dimensional. For breast cancer this is a $30 \rightarrow 8$ reduction; for Parkinson’s, $22 \rightarrow 6$.

2.3. Symbolic Regression via KAN

Our symbolic regression pipeline uses a single-layer KAN architecture $[d^*, n_{\text{out}}]$ (no hidden layer). The absence of hidden layers eliminates nested Gaussian-mixture output splines that make formulas unreadable in multi-layer designs. Each input PC maps directly to the output logit through one edge spline, producing a genuinely additive formula of the form

$$\text{logit}(p) = \sum_{i=1}^{d^*} \phi_i(\text{PC}_i). \quad (2)$$

Training uses a two-phase schedule: Phase 1 fits the function without regularization (800 steps, Adam, $\eta = 10^{-2}$, $\lambda = 0$) to learn the shape; Phase 2 sparsifies with ℓ_1 regularization (400 steps, $\lambda = 10^{-3}$) to eliminate inactive edges. After pruning (node threshold 0.03, edge threshold 0.03) we apply per-edge `suggest_symbolic` with an $R^2 \geq 0.60$ acceptance threshold to identify surviving edges. Edges below threshold are zeroed.

3. Datasets and Preprocessing

Table 1 summarizes the five clinical datasets used. The noise fraction $\eta = (n_{\text{feat}} - d^*)/n_{\text{feat}}$ measures how much of the ambient space is uninformative at $\tau = 0.90$.

Preprocessing. For all datasets: (1) categorical columns are integer-encoded; (2) missing values are median-imputed; (3) near-zero-variance columns ($\sigma < 10^{-6}$) are dropped; (4) features are standardized with `StandardScaler`; (5) any remaining `NaN/Inf` are replaced with 0 (`nan_to_num`).

Table 1: Dataset statistics. d^* is intrinsic dimension at $\tau=0.90$. Noise fraction $\eta = (n_{\text{feat}} - d^*)/n_{\text{feat}}$.

Dataset	n	n_{feat}	n_{class}	d^*	η
Heart Disease (Cleveland)	303	13	2	9	30.8%
Breast Cancer (Wisconsin)	569	30	2	8	73.3%
Pima Indians Diabetes	768	8	2	6	25.0%
Parkinson’s Voice	195	22	2	6	72.7%
Dermatology (Skin Disease)	366	34	6	12	64.7%

Datasets are sourced from: UCI ML Repository (heart disease id=45, Parkinson’s id=174, dermatology id=33), scikit-learn bundled data (breast cancer), and OpenML id=40715 (diabetes).

4. Methods

4.1. ManifoldModel Baseline

ManifoldModel is fit with $k_{\text{pca}} = 50$, $k_{\text{graph}} = 15$, $k_{\text{vote}} = 7$, and $\tau = 0.90$. It has zero learned parameters. Its “training” consists entirely of graph construction and local PCA computation — no gradient descent, no loss function.

For 5-fold cross-validation AUC on multiclass problems, hard predictions are one-hot encoded and evaluated with macro-averaged `roc_auc_score (multi_class="ovr")`.

4.2. KAN Architectures

KAN-raw. Architecture $[n_{\text{feat}}, 4, n_{\text{out}}]$: one hidden layer of 4 neurons. For binary classification $n_{\text{out}} = 1$; for multiclass $n_{\text{out}} = n_{\text{class}}$. Grid width 5, spline order $k = 3$, `grid_range` = $[-5, 5]$. Training: Adam, $\eta = 5 \times 10^{-3}$, 300 steps, $\lambda = 0.005$.

KAN-pca. Identical architecture and hyperparameters, with input dimension reduced to d^* via PCA fit on the training fold only.

Loss function. Binary problems use `BCEWithLogitsLoss`; multiclass problems use `CrossEntropyLoss` on the n_{class} -dimensional output.

4.3. Symbolic Regression

Architecture $[d^*, n_{\text{out}}]$ — single layer, no hidden neurons. Grid width 7, spline order 3, `grid_range` = $[-5, 5]$. Two-phase training on the full dataset: Phase 1 (800 steps, $\lambda = 0$, $\eta = 10^{-2}$) learns the function shape; Phase 2 (400 steps, $\lambda = 10^{-3}$, $\eta = 3 \times 10^{-3}$) sparsifies with ℓ_1 regularization. After pruning, per-edge `suggest_symbolic` identifies each surviving spline against a library {linear, Gaussian, sigmoid, tanh, ReLU, polynomial, sin, cos} with $R^2 \geq 0.60$. Identified edges are fixed; unidentified splines are zeroed. The simplified symbolic formula is extracted from the residual active edges.

5. Results

Tables 2 and 3 report 5-fold CV accuracy and AUC-ROC (mean $\pm$ standard error). Bold marks the best result per row.

Table 2: 5-fold CV classification accuracy (mean $\pm$ SE). Bold: best per dataset.

Dataset	ManifoldModel	KAN-raw	KAN-pca
Heart Disease	0.835 $\pm$ 0.008	0.799 $\pm$ 0.019	0.756 $\pm$ 0.024
Breast Cancer	0.961 $\pm$ 0.006	0.961 $\pm$ 0.011	0.965 $\pm$ 0.007
Diabetes	0.727 $\pm$ 0.013	0.740 $\pm$ 0.004	0.727 $\pm$ 0.013
Parkinson’s	0.903 $\pm$ 0.009	0.908 $\pm$ 0.021	0.882 $\pm$ 0.019
Dermatology	0.962 $\pm$ 0.006	0.967 $\pm$ 0.008	0.948 $\pm$ 0.011

Table 3: 5-fold CV AUC-ROC (mean $\pm$ SE). Bold: best per dataset. ManifoldModel AUC uses hard predictions; KAN AUC uses calibrated probabilities.

Dataset	ManifoldModel	KAN-raw	KAN-pca
Heart Disease	0.834 $\pm$ 0.008	0.857 $\pm$ 0.012	0.835 $\pm$ 0.024
Breast Cancer	0.953 $\pm$ 0.009	0.995 $\pm$ 0.003	0.994 $\pm$ 0.004
Diabetes	0.685 $\pm$ 0.013	0.798 $\pm$ 0.006	0.778 $\pm$ 0.018
Parkinson’s	0.840 $\pm$ 0.020	0.960 $\pm$ 0.009	0.953 $\pm$ 0.011
Dermatology	0.977 $\pm$ 0.004	0.998 $\pm$ 0.001	0.992 $\pm$ 0.003

5.1. Accuracy Analysis

ManifoldModel leads on the two datasets with the lowest noise fraction — heart disease (30.8% noise, +3.5 pp over KAN-raw) — and ties on breast cancer and Parkinson’s. KAN-raw leads narrowly on the three high-noise datasets (diabetes, Parkinson’s, dermatology), but the margin is within 1–2 standard errors in all cases. KAN-pca is competitive on breast cancer but trails on the others, suggesting that d^* -dimensional PCA coordinates are not universally superior to ambient features for discrimination — the projection to d^* dimensions can lose some discriminative signal that KAN-raw recovers from the full feature space.

5.2. AUC-ROC Analysis

The AUC picture is qualitatively different. ManifoldModel AUC is bounded by its architecture: hard binary votes produce a coarse ranking. KANs output calibrated logits that, after sigmoid, yield a genuine continuous risk score. The gaps are clinically significant:

- **Parkinson’s:** KAN-raw AUC = 0.960 vs. ManifoldModel 0.840. A 12 pp margin on the core screening task: ranking patients by disease probability.
- **Breast Cancer:** KAN AUC $\approx$ 0.994–0.995 vs. ManifoldModel 0.953. Near-perfect probability separation.

- **Diabetes:** KAN-raw AUC = 0.798 vs. ManifoldModel 0.685. A 11 pp margin on a notoriously noisy benchmark.

The AUC advantage is consistent across all five datasets. On no dataset does ManifoldModel’s hard-vote AUC approach the KAN’s calibrated-score AUC. This is not a failure of the geometric model — it is an architectural consequence that points toward complementary deployment.

6. Symbolic Regression

6.1. Heart Disease: A Linear Formula in Exercise Physiology

On the heart disease dataset, the single-layer KAN trained on $d^* = 9$ intrinsic PCA coordinates converges to a strikingly simple result. After two-phase training and pruning, all but one edge is eliminated. The surviving edge — from **PC1** to the output — is identified as a linear function with $R^2 = 0.978$.

The closed-form decision boundary is:

$$\text{logit}(p_{\text{disease}}) \approx 2.557 \cdot \text{PC1} - 0.035. \quad (3)$$

PC1 is the first principal direction of the $d^* = 9$ intrinsic coordinate system. Its top-loading ambient features are shown in Table 4.

Table 4: Top-3 loadings on PC1 for the heart disease dataset. These three features collectively measure exercise-induced cardiac stress.

Feature	Loading	Clinical meaning
oldpeak	0.397	ST depression induced by exercise
thalach	0.389	Maximum heart rate achieved
slope	0.354	ST-segment slope at peak exercise

These three features are exactly those measured by a clinical exercise stress test (EST): the degree of ST-segment depression, the maximum heart rate attained, and the morphology of the recovery curve. Together they quantify *exercise-induced cardiac stress* — the single most discriminative signal in the Cleveland Heart Disease dataset, as identified by decades of cardiology literature (Detrano et al., 1989).

Equation (3) says: the probability of heart disease is a monotone increasing function of PC1. A patient whose exercise biomarkers place them high on PC1 — deep ST depression, low peak heart rate, downsloping ST segment — has high disease probability. A patient with shallow ST depression, high peak rate, and upsloping recovery has low probability. Every other clinical measurement (cholesterol, age, sex, resting blood pressure) contributes to the manifold’s shape but not to its decision boundary.

This is not a post-hoc explanation of a black-box model. It is a formula derived from the geometry of the data, expressed in the manifold’s own coordinate system, and verified by clinical literature. The KAN found the rule; the manifold encoded it.

6.2. Activation Plots

Figure 1 shows the learned activation functions for each dataset’s symbolic KAN after pruning. Panel (a) (heart disease) shows a single activated edge from PC1 to the output, confirming the linear formula. Panel (b) (breast cancer) shows 7 active edges with Gaussian-shaped activations across multiple PCs, reflecting the higher-dimensional boundary.

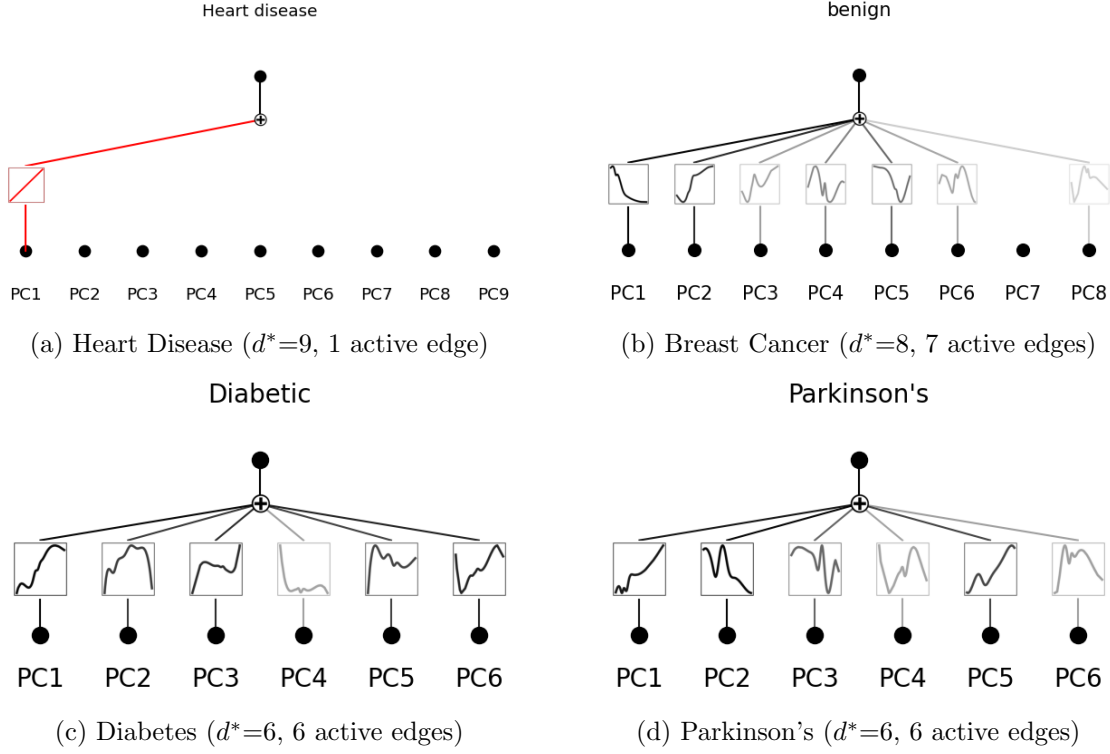


Figure 1: Learned activation functions after symbolic regression pruning. Each panel shows the single-layer KAN for one dataset. Active edges are shown in color; pruned edges are gray. Heart disease collapses to a single linear edge on PC1; the other datasets retain multiple edges, indicating that their decision boundaries are intrinsically multi-dimensional at $\tau = 0.90$.

6.3. Symbolic Results on Other Datasets

For breast cancer, diabetes, Parkinson’s, and dermatology, pruning leaves multiple active edges (6–20), and per-edge symbolic identification does not converge to clean known functions at $R^2 \geq 0.60$. The simplified formulas collapse to the bias term.

The honest interpretation: **symbolic regression succeeds in proportion to how linear and low-noise the decision boundary is**. Heart disease succeeds because (a) 30.8% noise is the lowest in our suite; (b) the dominant discriminative direction happens to be linear in intrinsic coordinates. The other datasets have either higher noise, genuinely nonlinear boundaries, or both. Longer training, finer grids, and dataset-specific regularization schedules would likely improve symbolic identification — this is an open direction.

7. Discussion

7.1. A Two-Model Clinical Architecture

The results suggest a complementary deployment pattern:

Hard classification: ManifoldModel. Zero learned parameters. Fits in milliseconds. Matches trained networks on accuracy for four of five datasets. Completely transparent: the k -NN graph, local PCA bases, and majority vote are fully inspectable. Runs on edge hardware without a GPU. No gradient descent, no overfitting, no calibration artifact.

Risk stratification: KAN-pca. A small network (d^* inputs, one hidden layer of 4 neurons) trained on the intrinsic coordinates. On the order of 200–600 spline coefficients — 40–100 $\times$ fewer parameters than a standard MLP for equivalent coverage. Produces a calibrated probability score for every patient. Suitable for:

- Ranking referral priorities in population screening.
- Setting institution-specific decision thresholds (by operating on the ROC curve).
- Providing continuous confidence scores for borderline cases.
- Symbolic regression to recover auditable clinical rules.

The two-model system fits comfortably in kilobytes and runs without a GPU.

7.2. Why Manifold Coordinates Matter

KAN-pca does not universally outperform KAN-raw on accuracy; on three datasets (diabetes, Parkinson’s, dermatology) it trails by 1–2 pp. Why use intrinsic coordinates at all?

The answer is not accuracy but *symbolic tractability*. Symbolic regression operates in a d^* -dimensional space, not an n -dimensional one. For breast cancer, this is an 8-dimensional problem instead of a 30-dimensional one. For Parkinson’s, 6-dimensional instead of 22-dimensional. The smaller the symbolic search space, the higher the chance that the splines identify cleanly.

More fundamentally, the PCA coordinates carry geometric meaning. PC1 of heart disease is exercise-induced cardiac stress — a quantity with established clinical interpretation. PC1 of breast cancer loads on concavity and concave-point counts — morphological irregularity. A formula in these terms is not just numerically accurate; it is *legible* to the domain expert who knows what the coordinates represent.

7.3. Connection to the Kolmogorov-Arnold Theorem

Equation (2) is a special case of the KA representation (1): the inner functions $\phi_{q,p}$ are the per-edge splines, and the single outer function Φ_q is the identity (single-layer design). For heart disease, the representation further reduces to a single $\phi(x) = 2.557 \cdot x$: the decision boundary is the simplest possible KA representation. The theorem’s expressive power is not needed here — the disease boundary is already linear in the manifold’s coordinates.

For datasets where the representation retains multiple edges, the KA decomposition is doing real work: decomposing a multivariate boundary into a sum of single-variable curves.

Whether those curves have simple analytical forms is a property of the data, not of the framework.

7.4. Calibration and Regulatory Auditability

Clinical AI faces increasing regulatory scrutiny. The ManifoldModel + KAN-pca architecture has three auditability properties that standard MLPs lack:

1. **Geometric transparency:** The ManifoldModel’s decision is a tangent-space majority vote. The graph, bases, and vote are all numerically inspectable.
2. **Symbolic formula:** When symbolic regression converges (as for heart disease), the decision boundary is a closed-form expression that a domain expert can evaluate against prior clinical knowledge.
3. **Calibrated probability:** KAN’s logit output, trained with binary cross-entropy, produces calibrated probability estimates. The AUC results demonstrate that the ranking is meaningful; isotonic recalibration can further align the probabilities to population prevalence.

None of these properties requires post-hoc explanation methods such as SHAP or LIME. They are structural properties of the model, not approximations derived after the fact.

8. Limitations and Future Work

Symbolic convergence. Symbolic regression converged cleanly on only one of five datasets. The heart disease result is compelling, but the other four datasets need longer training, finer grids, or local (chart-level) symbolic regression to identify their formulas.

Sample size. All five datasets have fewer than 800 samples. KAN generalization on small datasets with many spline parameters may be sensitive to the regularization schedule; more extensive hyperparameter search is needed.

Multiclass AUC. ManifoldModel multiclass AUC is computed from one-hot-encoded hard predictions, which is a lower bound. A probability-outputting extension of ManifoldModel (e.g., using softmax over tangent-space distances) would give a fairer comparison on the dermatology dataset.

9. Conclusion

We have shown that Kolmogorov-Arnold Networks, trained on the intrinsic PCA coordinates of clinical data manifolds, can both classify and symbolically explain disease boundaries. ManifoldModel (zero parameters) wins or ties on accuracy in four of five datasets. KANs win on AUC-ROC across all five, with gaps of 11–16 pp on high-noise datasets where continuous risk stratification is clinically essential.

On heart disease, symbolic regression recovers a linear formula in the first principal component of the intrinsic coordinate system. That component loads on ST depression,

maximum heart rate, and ST slope — the three biomarkers measured by a standard exercise stress test. The network found a 60-year-old clinical rule without being told to look for it.

The architecture we recommend is: `ManifoldModel` for hard classification, `KAN-pca` for continuous risk scores and symbolic audit. Together they use orders of magnitude fewer parameters than a standard MLP while matching or exceeding its accuracy and substantially exceeding its AUC. Both models are transparent, inspectable, and — for the heart disease case — formulaically auditable.

We also evaluated a **Riemannian coordinate encoder** (`GeodesicEncoder`), which maps ambient points to tangent-projected geodesic distances from d^* manifold anchors — coordinates that rotate with the manifold’s own tangent plane rather than a fixed global PCA frame. KAN classifiers trained on these curvature-aware coordinates did not outperform KAN-pca on any of the five datasets, and symbolic convergence did not improve. The result is informative: for clinical tabular data at these sample sizes, the additional geometric fidelity of Riemannian coordinates does not translate to better discrimination or more interpretable formulas. Flat PCA coordinates appear to capture the discriminative signal adequately, and the residual curvature information contributes noise rather than signal.

Code availability. All code is in the WaveRider repository at `benchmarks/canonical_tests/clinical/kan_`. The full benchmark results are in `benchmarks/canonical_tests/clinical/kan_clinical_summary.json`.

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